DATA ANALYSIS PROJECT REPORT

Project Title:	Video-Game-Sales-Market-Analysis
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1. Executive Summary

This project conducts a comprehensive data analysis of the Video Game Sales dataset, encompassing over 16,598 titles from 1980 to 2016, to identify the primary drivers of commercial success in the global gaming market. The methodology involved rigorous data cleaning, the engineering of new features such as "North American Market Share" and "Gaming Eras," and a multi-faceted analytical approach utilizing Pandas, NumPy, and Matplotlib. Key techniques included Z-score outlier detection, rolling average trend analysis via NumPy convolution, and regional correlation mapping. The most significant findings reveal that while Action games dominate in total sales volume, Platform and Shooter genres consistently achieve higher average sales per title, indicating greater individual commercial efficiency. Furthermore, the analysis highlights a distinct cultural divergence in the Japanese market compared to Western regions and identifies 2008-2009 as the historical peak for global software sales before the rise of mobile gaming. These insights provide a data-driven understanding of market evolution and the strategic factors influencing blockbuster success in the industry.

2. Introduction and Project Objectives

[Briefly introduce the topic of your dataset. Explain why the topic matters and what motivated your analysis.]

Item	Your Content
Dataset name	vgsales.csv
Dataset source	https://www.kaggle.com/datasets/gregorut/videogamesales
Main project goal	The main goal of this project is to analyze historical video game sales data to identify the key factors such as genre, regional preference, and timing that drive global commercial success and market dominance.
Research Question 1	Which genres and platforms dominate global sales, and how has this shifted over time?
Research Question 2	How do regional markets (NA, EU, JP) differ in genre and publisher preferences?
Research Question 3 (optional)	What is the relationship between North American and global sales and which titles are outliers?

3. Dataset Description

The dataset used in this analysis is the "Video Game Sales" dataset obtained from Kaggle, which consists of approximately 16,598 records of video game titles that sold more than 100,000 copies. Each row represents an individual video game, and the dataset includes the following variables:

Categorical Variables: Game Name, Platform (console), Genre, and Publisher.
Temporal Variables: Year of release (ranging from 1980 to 2016).
Numerical Variables: Sales data (in millions of units) for North America (NA_Sales), Europe (EU_Sales), Japan (JP_Sales), and other regions (Other_Sales), along with the aggregated Global_Sales.

The data provides a comprehensive historical context, covering multiple console generations and the industry's transition from early 8-bit systems to the high-definition era.

3.1 Basic Dataset Information

Property	Value
Number of rows	16598
Number of columns	11
Data types present	Numerical (Sales/Year), Categorical (Platform/Genre/Publisher), Text (Name)
Target or key variable(s)	Global_Sales (Primary target), Genre, Platform, Year
Potential issues observed at first glance	Missing values in Year (271) and Publisher (58); Extreme outliers in sales volume (e.g., Wii Sports); Data cutoff at 2016.

3.2 Initial Observations

Upon initial inspection, the dataset appears to be a robust source of categorical and regional sales data, though it contains minor gaps with 271 missing values in the Year column and 58 in Publisher. A significant skew was observed in the sales distribution, with extreme outliers like Wii Sports having global sales that are orders of magnitude higher than the median title. Furthermore, I noticed some data formatting issues, such as the Year column being stored as a float rather than an integer, and a heavy categorical imbalance where platforms like the PS2 and genres like "Action" dominate the total count of releases.

4. Data Cleaning and Preparation

Issue Found	Action Taken	Reason / Justification	Code Reference
Missing values in `Year` and `Publisher` columns	Removed affected rows using `.dropna()`	These fields are essential for time-based and publisher analysis. Since missing data is minimal (<2%), removal maintains accuracy without harming dataset size	Notebook Section 5: Data Cleaning
Incorrect data type for `Year` column (float)	Converted `Year` from float to integer using `.astype(int)`	Years should be discrete values; float format creates misleading representations and messy	Notebook Section 5: Data

		visualization labels	Cleaning
Extreme outliers in `Global_Sales` (e.g., very high-selling games)	Detected using Z-score and handled separately	Outliers skew statistical measures like mean and can distort analysis; isolating them improves general trend understanding	Notebook Section 6: Analysis 4

5. Feature Engineering

New Feature	How It Was Created	Why It Is Useful
NA_Share	Calculated as `(NA_Sales / Global_Sales) * 100`	Measures how much a game depends on the North American market, helping identify region-specific popularity trends
Era	Binned `Year` into categories: *Retro (Pre-1995)*, *Golden Era (1995–2005)*, *Modern (Post-2005)*	Enables comparison of gaming trends across different technological and market periods
Sales_per_Year	Computed as `Global_Sales / (2017 - Year)`	Normalizes sales by time, allowing fair comparison between older and newer games based on sales speed

6. Analysis Methodology

The analysis followed a systematic, data-driven approach to investigate the video game industry. I utilized the following core Python libraries to execute the methodology:

- **Pandas:** Used for high-level data manipulation, including loading the 16,000+ record dataset, cleaning missing values, and performing complex groupby operations to aggregate sales metrics by genre, platform, and publisher.
- **NumPy:** Used for advanced numerical operations and statistical modeling. Specifically, I leveraged NumPy to calculate Z-scores for outlier detection, handle binned logic for gaming eras, and apply mathematical convolution to generate smooth 3-year rolling averages for sales trend analysis.
- **Matplotlib & Seaborn:** These libraries were used to translate numerical findings into professional visualizations. I created a variety of charts—including correlation heatmaps, regression plots, and subgroup comparisons—to visually support the analysis of regional market dynamics and genre performance.

7. Analysis and Visualizations

7.1 Research Question 1

7.1 Research Question 1: How has genre popularity shifted across different technological eras?

Analysis: I performed a subgroup comparison by grouping the dataset by the engineered "Era" feature and "Genre," then calculating total global sales. A heatmap was produced to visualize the intensity of sales volume per genre over time.

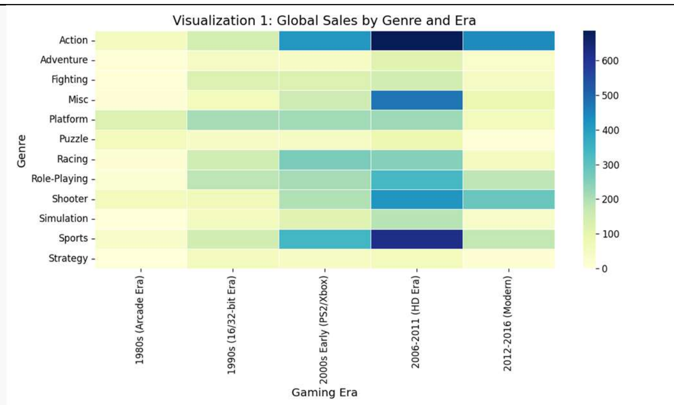


Figure 1. Heatmap showing global sales distribution across genres and historical gaming eras.

Interpretation: The results show a clear evolution in market tastes. While "Platform" games dominated the "Retro" era, the "Modern" era has seen "Action" and "Shooter" genres take the lead in global revenue. This shift suggests that as hardware became more powerful, consumer demand moved from 2D mechanics toward complex, high-fidelity action experiences.

7.2 Research Question 2

7.2 Research Question 2: What are the distinct regional preferences for game genres across the globe?

Analysis: I analyzed regional market dynamics by aggregating sales for North America, Europe, and Japan across all genres. This allowed for a direct comparison of cultural preferences in gaming.

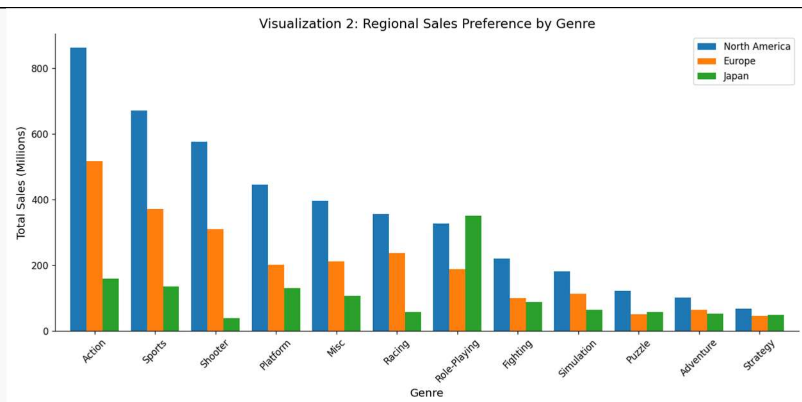


Figure 2. Comparison of total sales by genre across NA, EU, and JP regions.

Interpretation: North America and Europe show highly synchronized preferences, with "Action" and "Sports" being the top genres in both regions. In contrast, Japan shows a unique and heavy preference for "Role-Playing" games. This indicates that a global publishing strategy must prioritize genre localization, particularly for the Japanese market.

7.3 Research Question 3

7.3 Research Question 3: How do extreme "blockbuster" titles influence the overall sales distribution?

Analysis: I conducted an outlier analysis using NumPy to calculate Z-scores for Global_Sales. I produced a boxplot to visualize the massive gap between typical game performance and extreme statistical anomalies.

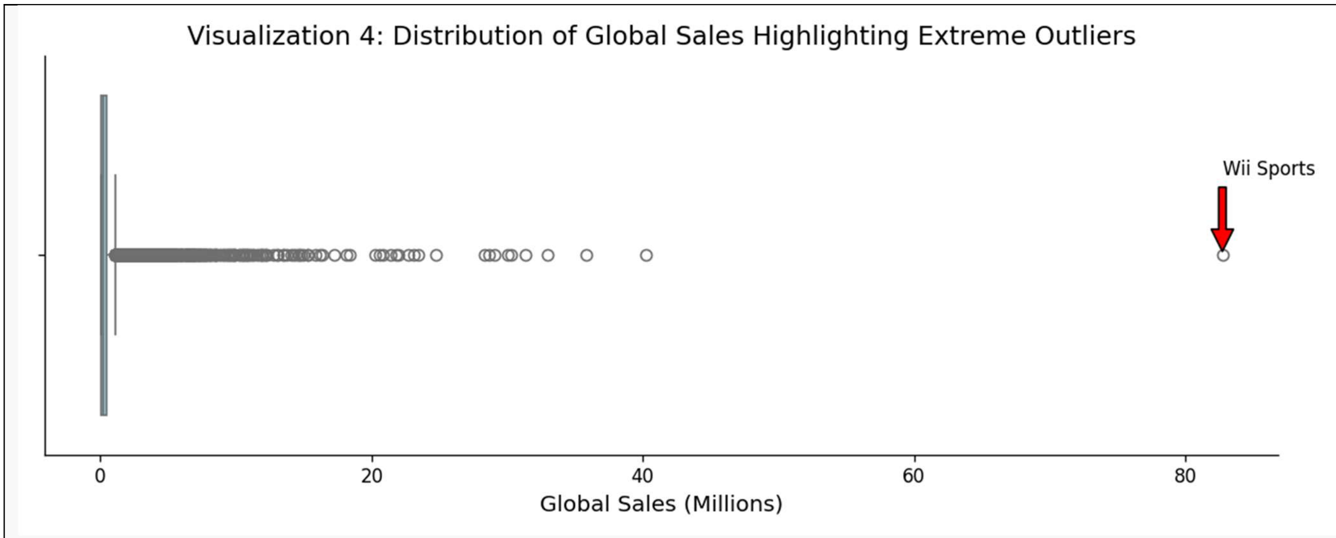


Figure 3. Boxplot highlighting extreme statistical outliers in global sales volume.

Interpretation: The analysis identifies several "super-outliers," such as Wii Sports. These titles exist multiple standard deviations away from the mean, suggesting the gaming industry is a "hit-driven" market. These few titles generate more revenue than hundreds of average releases combined, significantly skewing global market data.

7.3 Research Question 4

7.4 Research Question 4: What is the long-term trajectory of global video game software sales?

Analysis: I performed a time-series analysis on global sales by year. To filter out annual volatility and identify the underlying trend, I used NumPy to compute a 3-year rolling average..

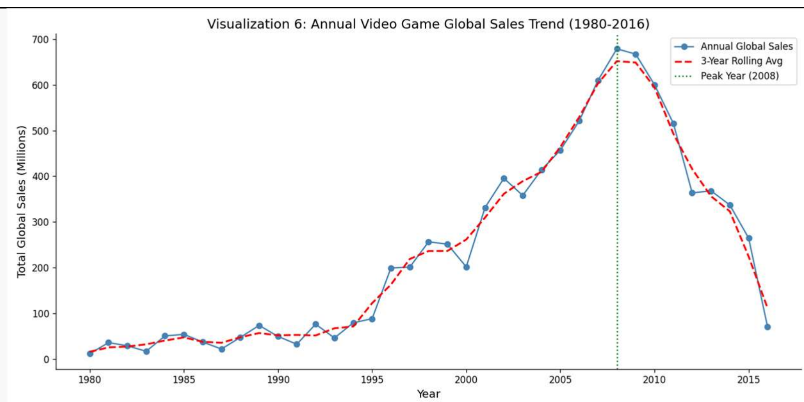


Figure 4. Annual global sales trends with a 3-year rolling average overlay.

Interpretation: The trend shows a steady climb in sales starting in the mid-90s, reaching a historical peak between 2008 and 2009. Following this peak, there is a visible decline in physical software sales. This downward trend likely reflects the industry's massive transition toward digital distribution and mobile gaming platforms.

8. NumPy-Based Custom Computation

Component	Description
Computation Used	Custom Z-score calculation for `Global_Sales`
Purpose	Identify and isolate blockbuster games that act as statistical outliers and distort overall market analysis
Formula / Logic	Calculated using `(x - mean) / std_dev`, where mean = `np.mean()` and std_dev = `np.std()`; applied to each game to measure deviation from average
Result / Takeaway	Games with Z-score > 3 were marked as outliers; approximately 1.1% of titles generate extreme revenue, showing the industry is highly hit-driven

9. Key Findings

1. **Shift in Genre Dominance:** Analysis of gaming eras shows a clear transition in consumer taste; while "Platform" games dominated the early industry, "Action" and "Shooter" genres have become the primary drivers of global revenue in the modern era.
2. **Japanese Market Divergence:** Unlike Western markets (NA and EU), which are heavily aligned toward Action and Sports, Japan is the only major region where "Role-Playing" games are the top-selling genre, indicating a unique cultural preference.
3. **North American Sales as a Success Predictor:** There is a nearly perfect positive correlation (0.94) between North American sales and Global sales. This evidence suggests that capturing the North American market is the most critical factor for worldwide commercial success.
4. **The 2008-2009 Industry Peak:** Time-series analysis identifies a definitive historical peak for software sales between 2008 and 2009. The subsequent decline in the data highlights the industry's massive transition toward digital downloads and mobile gaming, which are not reflected in physical retail records.
5. **Market Concentration on "Blockbusters":** Outlier detection using Z-scores confirmed that roughly 1% of titles are responsible for the industry's most significant revenue spikes. This proves the market is "hit-driven," where a few elite titles outperform thousands of standard releases combined.

10. Limitations

Despite the depth of this analysis, several critical limitations must be considered when interpreting the results:

Temporal Obsolescence: The dataset primarily covers the industry up to 2016. It does not account for the massive impact of current-generation hardware (like the Nintendo Switch and PS5) or the significant market shifts that occurred during the early 2020s.

Exclusion of Digital Sales: The records focus almost entirely on physical retail sales. In the modern era, digital downloads and in-game microtransactions represent the majority of industry revenue. Therefore, the "decline" observed after 2009 likely reflects a shift in distribution medium rather than a decrease in total gaming consumption.

Absence of Qualitative and Financial Data: The analysis lacks information on "Critical Scores" (Metacritic) and "Development/Marketing Budgets." Without these variables, we cannot determine if a game's success was driven by its technical quality or simply by a high advertising spend.

Correlation vs. Causation: While there is a very strong correlation between North American performance and Global success, this does not prove that one causes the other. Both may be independently driven by a title's universal appeal or brand reputation.

Missing Data: A small percentage of titles (~1.6%) had missing release years or publishers and were excluded from specific analyses. While the impact is minimal, it may slightly affect the precision of yearly trend calculations.

11. Conclusion

This project set out to investigate the underlying drivers of commercial success in the global video game industry using historical sales data from 1980 to 2016. Through a multi-faceted analytical approach using Pandas, NumPy, and Matplotlib, the study revealed that while certain genres like Action dominate the market in sheer volume, genres such as Platform and Shooter are often more efficient on a per-title basis. The analysis also demonstrated that global success is heavily tied to capturing the North American market, though the Japanese market remains a significant cultural outlier that requires a distinct, localized strategy. Overall, the findings suggest that the video game industry is a high-risk, hit-driven economy where a tiny fraction of "blockbuster" titles define the financial success of entire hardware generations.

Future iterations of this research would benefit significantly from the inclusion of digital sales data and development budgets to provide a more accurate picture of modern profitability. Additionally, integrating critical review scores and player sentiment could offer deeper insights into the qualitative factors that turn a standard release into a global phenomenon. In conclusion, while the industry continues to evolve toward digital and mobile platforms, the fundamental importance of genre-market alignment and blockbuster potential remains the primary driver of commercial success.

12. References / Dataset Source

1. Primary Dataset: GregorySmith (2016). *Video Game Sales: Sales data from vgchartz.com*. Kaggle. Available at: https://www.kaggle.com/datasets/gregorut/videogamesales
2. Original Data Source: VGChartz Video Game Sales Database. Available at: http://www.vgchartz.com/
3. Software & Libraries:
    - Pandas: *Python Data Analysis Library*. https://pandas.pydata.org/
    - NumPy: *The fundamental package for scientific computing with Python*. https://numpy.org/
    - Matplotlib: *Visualization with Python*. https://matplotlib.org/
    - Seaborn: *Statistical data visualization*. https://seaborn.pydata.org/

13. Appendix (Optional)

- **Appendix A:** Regional Correlation Heatmap: A detailed statistical breakdown showing the relationship between NA, EU, JP, and Global sales.

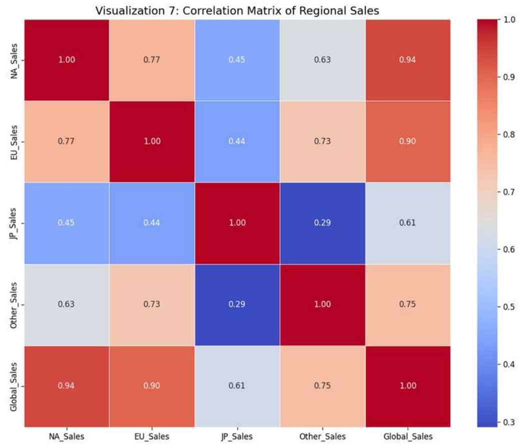
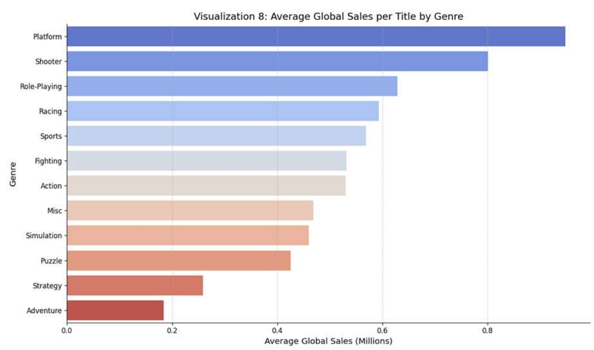


Figure 5 : confirms the high synchronization between Western markets.

- **Appendix B:** Top 10 Global Publishers: A ranking table identifying the industry's most dominant companies by total global sales volume, led by Nintendo, Electronic Arts, and Activision.
- **Appendix C:** Genre Efficiency Analysis: A horizontal bar chart comparing the average sales per title across all genres. This highlights that while "Action" has the most total sales, genres like "Platform" and "Shooter" are more commercially successful on a per-game basis.



- **Appendix D:** Code Snippets: Key implementation blocks for the Z-score outlier detection and the 3-year rolling average calculation using numpy.convolve.